ONLINE APPENDIX

# Dissecting LLM News Sentiment Scores: De-Identification, Batch Composition, Scoring Noise, and Price Echoes in Return Predictability

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Companion to the main paper. This document contains Appendices A–K: the scoring prompt, the anonymization procedure and its audit, firm-level coverage, auxiliary analyses, the power and equivalence tables, the staged-session design documents with their prediction–outcome tables (sessions 3–6b), the mechanism and downstream accounting of the batch-composition effect, the classification audit of the price-language flag, and the minimal audit protocol. Section, table, and figure numbers cited as bare numbers refer to the main paper; letters (A–K) refer to sections of this appendix.

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## A LLM Scoring Prompt

The following system prompt was applied verbatim to every batch of 25 headlines in all conditions; the named and repeat conditions prefix each headline with "[Company]", and the anonymized condition presents the de-identified headline text with no prefix. The scoring model is `glm-4-plus` (Z.ai), accessed through its API. Every request disables the model's extended-reasoning ("thinking") mode —a request-level switch that turns off chain-of-thought generation before the answer, forcing direct output — and otherwise uses the provider's default decoding parameters: no temperature, top-$p$, or seed overrides, so the sampling distribution is whatever the endpoint defaults to, and its stochasticity is a measured property of the instrument (Section 5.3) rather than an assumption. Batches that failed to parse were retried up to three times. All raw responses and batch logs are archived in the replication package.

```
You are an expert financial analyst specialized in news sentiment analysis for
equity research. You evaluate how news headlines affect a company's stock value.

For each headline you must return:
- "sentiment": integer from -100 to +100. -100 = extremely negative impact on the
company's stock value; 0 = neutral or no clear impact; +100 = extremely positive
impact. Judge from the perspective of a shareholder of the company.
- "relevance": integer from 0 to 100. 100 = headline is directly about the company's
business, earnings, products, guidance, or stock; 0 = headline mentions the company
only incidentally or is a generic market/quote page.

Rules:
- Base your judgment ONLY on the information in the headline text.
- Earnings beats, raised guidance, product launches, upgrades, buybacks = positive.
- Earnings misses, cut guidance, lawsuits, recalls, downgrades, layoffs = negative.
- Speculation or rumor framing lowers the magnitude toward zero.
- Generic quote pages, price summaries without direction, or calendar pages =
sentiment 0, relevance low.

Return ONLY a valid JSON array. One object per headline, in the same order as given,
with fields: "id", "sentiment", "relevance". No other text.
```

## B Anonymization Procedure

Company identifiers are removed by a case-insensitive alias table mapping each firm's name, common short forms, and ticker to the phrase "the company" (e.g., "Apple Is Up 20% in 2026. What Will It Take for AAPL Stock" becomes "the company Is Up 20% in 2026. What Will It Take for the company Stock"). 325 of 341 in-window headlines (95.3%) are altered. The procedure is conservative: product names, third-party firms mentioned inside a headline, and stylistic cues are left intact, so the anonymized condition is a lower bound on de-identification. Quantifying the residual firm-identity component under the executed alias table: 5 of 341 anonymized headlines (1.5%) still name at least one other sample firm —under both a case-insensitive and a case-sensitive matching convention —with analyst actions ("BofA," "Morgan Stanley") and co-mentioned mega-caps the typical cases. The dominant de-identification gap is therefore product language (e.g., device or franchise names) rather than residual firm names. Ambiguous tickers that collide with common words (e.g., "COST," "CAT," "V") are replaced only in their exact all-caps form.

**A disclosed failure of the procedure.** The executed alias table contains no entry for ticker V (whose identifiers would have been "Visa"), and all ten in-window V headlines contain the word "Visa" but never the standalone all-caps token "V." The anonymization was therefore a no-op for those ten headlines: the text sent to the model under the anonymized condition still named the focal company —2.9% of the 341 —and the condition contrast for them is effectively named-versus-named. Five of the ten are U.S. State Department "Visa Bulletin" immigration headlines (unrelated to the payments company; all scored at relevance 20, so they never enter the firm-day sample but do enter the article-level tests), and five enter the relevance-qualified sample. The failure is instructive in two ways. First, it materially moves the marginal same-session identity contrast: excluding the ten V headlines sharpens it from $-1.04$ ($p = 0.11$) to $-1.29$ ($p = 0.04$; Wilcoxon $p = 0.03$), which is why Section 5.2 reports the contrast as a range rather than a verdict. Second, and more generally, it was caught only by re-running the anonymizer with a verification pass —checking that anonymized text no longer matches the focal firm's identifiers —which we now treat as a mandatory step of the protocol rather than an optional one. The illustrative pair: the named condition "[Visa] Visa Stock Just Posted Its Best Quarter Since 2022." scored +80 (relevance 90), while the "anonymized" condition —the identical text minus the prefix —scored 0; the difference is a clean single-headline demonstration of how much identity information the prefix-plus-name carries. Computed by `v_exclusion.py` from the archived corpus and scores.

## C Firm-Level Coverage

Table 1 reports news coverage by firm: in-window headlines and relevance-qualified trading-day firm-days. Coverage ranges from 4 to 17 articles per firm and correlates with firm prominence in web search, which motivates the selection analysis of Section 5.5 and the size interaction of Appendix D. The table is generated directly by `coverage_table.py` from the archived corpus, scoring, and panel files, so every cell is reproducible; the generator asserts that the rendered cells show each of the 36 firms exactly once and that the column totals equal the verified corpus totals (341 in-window articles, 240 relevance-qualified trading-day firm-days).

## D Auxiliary Analyses

This appendix reports the three descriptive exercises summarized in Section 6.3: the horizon path, the portfolio sorts, and the heterogeneity interactions. None is part of the declared test family; each is reported under the audit's inference standard.

**Table 1:** News coverage by firm (in-window articles and trading-day firm-days)

| Ticker | Articles | Firm-days | Ticker | Articles | Firm-days | Ticker | Articles | Firm-days |
|---|---|---|---|---|---|---|---|---|
| AAPL | 16 | 9 | GE | 8 | 6 | MSFT | 10 | 5 |
| ADBE | 10 | 9 | GOOGL | 6 | 4 | NFLX | 14 | 11 |
| AMD | 13 | 8 | GS | 11 | 7 | NVDA | 13 | 8 |
| AMZN | 8 | 6 | IBM | 8 | 7 | ORCL | 12 | 9 |
| AVGO | 13 | 10 | INTC | 14 | 11 | PFE | 5 | 4 |
| BA | 5 | 4 | JNJ | 7 | 4 | PG | 6 | 4 |
| BAC | 7 | 5 | JPM | 4 | 4 | QCOM | 7 | 7 |
| CAT | 6 | 5 | KO | 10 | 6 | TSLA | 9 | 8 |
| COST | 15 | 11 | LLY | 6 | 6 | V | 10 | 3 |
| CRM | 17 | 8 | MA | 9 | 8 | WFC | 12 | 9 |
| CVX | 7 | 5 | META | 11 | 7 | WMT | 13 | 9 |
| DIS | 4 | 4 | MS | 7 | 2 | XOM | 8 | 7 |

Articles: headlines dated 2026-07-01 to 2026-10-14 in the archived corpus (column total 341). Firm-days: unique firm–date cells with scored relevance $\geq 30$ falling on a trading date (column total 240, equal to the regression sample). 36 firm–date cells fall on non-trading dates and are dropped without roll-forward (Section 4). Computed by `coverage_table.py` from `news_all.json`, `llm_scores.json`, and `panel.csv`.

### D.1 The horizon path, re-read

The horizon path shows a sign pattern —positive at $t+1$ ($t = 2.10$), negative at $t+2$ ($t = -2.04$, $p = 0.042$), fading by $t + 3$ ($t = -1.52$) —that invites an attention-driven overshooting-and-correction reading. The audit re-reads the pattern rather than discarding it. A subsample selected on high absolute returns (H5) and scored with text that partially echoes those returns (H4) mechanically produces short-horizon continuation followed by reversal: this is the classical news-event return dynamics documented by Chan (2003) for reasons (liquidity provision, bid–ask bounce, order-flow imbalance) that require no sentiment semantics at all. Under the decomposition (1), the observed path is what $\rho > 0$ plus event-selection generates; the reversal is therefore *consistent with* the echo channel and does not independently identify overshooting. We accordingly read the reversal as a pattern consistent with the echo interpretation rather than as an independent finding, and note that the next-day open-to-close coefficient is small and insignificant ($t = 0.45$) —better read as the echo's asymmetry between the overnight and intraday windows on an underpowered sample than as evidence of overnight diffusion.

### D.2 Portfolio sorts, honestly reported

Sentiment-sorted quintile portfolios on the baseline sample earn −20.3, +18.9, +6.5, +35.0, and +69.6 bps from Q1 to Q5, a Q5−Q1 spread of 89.9 bps with a Welch $t = 1.24$ ($p = 0.22$). Three facts deserve equal prominence: the sort is *not monotone* (Q2 exceeds Q3); the spread is not statistically significant; and the IC and direction tests are null (mean daily IC +0.035, $t = 0.30$ over 25 usable cross-sections; direction logit $p = 0.20$, marginal effect 11.5 points). Under the audit's standard, the economic-significance evidence does not clear its own bar, and gross-of-cost spreads on an event-selected sample of 47 observations per quintile should not be quoted as tradable. No transaction-cost claim is made beyond the observation that any cost level above zero further erodes a statistically insignificant spread.

### D.3 Heterogeneity: interactions, not splits

A split-sample presentation —coefficient significance in above- versus below-median subsamples —is the natural first pass at heterogeneity, and it is the rhetoric the econometrics literature warns against, since separate significances do not test a difference. Table 2 reports the formal interaction-term tests. The size interaction is significant ($t = 2.25$, $p = 0.025$): predictability genuinely concentrates in above-median-dollar-volume firms, which in this corpus is nearly mechanical —mega-caps dominate search coverage, so the instrument is most precise (densest text input) exactly there. The news-intensity interac-

tion is marginal ($t = 1.90$, $p = 0.057$). The volatility interaction is zero ($t = 0.01$): a volatility-regime split showing coefficients that differ by 0.0007 across regimes with different standard errors implies no detectable interaction, so no state-dependence claim survives a formal test. Figure 1 contrasts the two inference styles directly —the point of the exercise is methodological as much as substantive.

**Table 2:** Heterogeneity: formal interaction tests versus split-sample statistics

| Channel | Interaction coefficient | $t$ | $p$ |
|---|---|---|---|
| Sentiment × large firm (above-median dollar volume) | 0.0248 | 2.25 | 0.025 |
| Sentiment × multi-headline firm-day | 0.0512 | 1.90 | 0.057 |
| Sentiment × high-volatility regime | 0.0001 | 0.01 | 0.994 |

Interactions from $r_{i,t+1} \sim$ sentiment + sentiment × modifier + modifier + same-day return + controls, two-way clustered, $N = 205$. A volatility-regime split ($t = 2.49$ vs. $t = 1.00$) implies no detectable interaction.

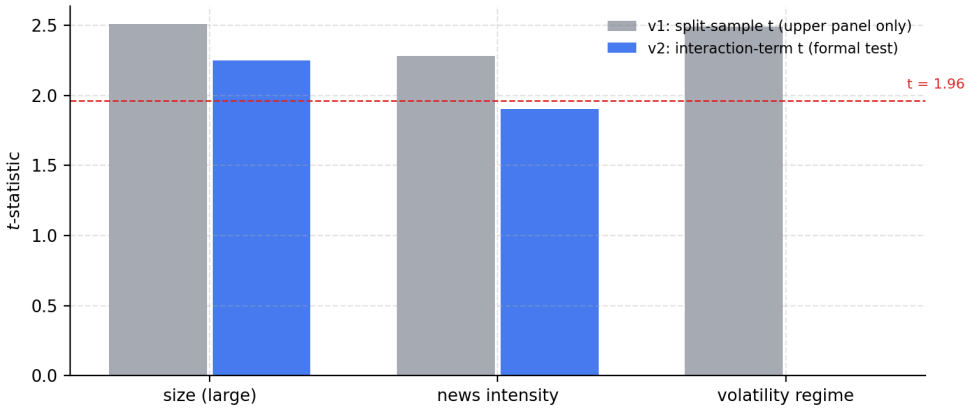



**Figure 1:** Split-sample $t$-statistics (grey) versus formal interaction $t$-statistics (blue) for the three heterogeneity channels. Only the size effect survives formal testing; the volatility claim collapses to zero.

# E Power and Minimal Detectable Effects

For each test in the declared family, Table 3 reports the observed effect, its standard error, the two-sided 5% minimum detectable effect at 80% power ($\text{MDE}_{80} = (z_{0.975} + z_{0.80}) \times \text{SE} = 2.80 \times \text{SE}$), and the power of the test against the observed effect, $\Phi(|\delta|/\text{SE} - 1.96) + \Phi(-|\delta|/\text{SE} - 1.96)$. Standard errors for regression coefficients are the two-way clustered values of Table 2; standard errors for the difference tests are implied by the bootstrap confidence intervals of Table 3 (width divided by $2 \times 1.96$); the level-shift rows use their paired standard errors; the selection row uses the two-way clustered standard error of the fixed-effects specification of Section 5.5, estimated on the $N = 2{,}628$ firm-days with non-missing same-day returns (the 2,664-cell panel less the 36 first-trading-day cells, which lack a prior close and hence a same-day return). The table quantifies what the paper's non-rejections can and cannot rule out: the named-versus-anonymized predictive difference rules out substitutions larger than 0.006 —about 42% of the baseline coefficient itself —while the named-versus-dictionary contrast rules out almost nothing ($\text{MDE}_{80} = 0.094$, more than six times the baseline). At the return layer, the design can only detect next-day coefficients of 0.020 or larger (72 bps per one-standard-deviation sentiment move, against the observed 54 bps), and the achieved power against the observed effect is 0.56: the marginal significance of H1 is exactly what an underpowered design produces around a true effect of this size. The H4 echo split has power 0.31 against its observed difference, which is why Section 5.4 labels it directional rather than established. At the measurement layer, the level-shift and selection tests are comfortably powered (0.999 and 0.79), which is the statistical counterpart of the paper's narrative: the channels the audit finds with confidence are the ones it can measure. The same-session identity row is the staged contrast: at $|\delta| = 1.04$ points and SE = 0.65, the design reaches power 0.36 —the marginal mean-test result is what a design of

this size produces around a one-point effect, and the rank test and V exclusion are the sharper instruments (Section 5.2).

**Table 3:** Power and minimal detectable effects for the declared family

| Test | Observed | SE | $\text{MDE}_{80}$ | Power |
|---|---|---|---|---|
| H1: baseline next-day coefficient (named) | 0.0149 | 0.0071 | 0.020 | 0.56 |
| H2: $\beta_{\text{named}} - \beta_{\text{anonymized}}$ | 0.0004 | 0.0023 | 0.006 | 0.05 |
| H2: $\beta_{\text{named}} - \beta_{\text{dictionary}}$ | −0.0119 | 0.0335 | 0.094 | 0.07 |
| H4: $\beta_{\text{price-lang.}} - \beta_{\text{clean}}$ | 0.0436 | 0.0297 | 0.083 | 0.31 |
| H2: identity level shift (points, $n = 341$) | 4.75 | 0.92 | 2.59 | 1.00 |
| H5: coverage selection (bps, $n = 2{,}628$) | 61.1 | 22.2 | 62.1 | 0.79 |
| H2: same-session identity shift (points, $n = 341$) | −1.04 | 0.65 | 1.83 | 0.36 |
| H2: interleaved identity shift (points, $n = 341$, session 3) | −2.82 | 1.03 | 2.88 | 0.78 |

Two-sided 5% tests; $\text{MDE}_{80} = 2.80 \times \text{SE}$. "Power" is achieved power at the observed effect. The identity-level-shift row is the cross-session contrast as declared; the same-session row is its staged re-estimate; the last row is the interleaved-session confirmatory contrast under the conservative batch-call-clustered SE (the headline-level SE of 0.66 would give power 0.99; the round-level meta-test is stronger still). The H4 and H5 rows use the $B = 9{,}999$ bootstrap CI and the fixed-effects clustered SE of the panel inference specification. Computed by `revision_power.py` (family and session-2 rows) and `analysis_v3.py` / `reinference.py` (session-3 row) from archived artifacts.

## E.1 Equivalence Assessment of the Instrument Contrasts

Table 3 runs the power logic one way —what differences this sample could detect. The instrument conclusions require running it the other way as well: a non-rejected difference is not evidence of equivalence, and several statements a reader might naturally form ("the conditions are indistinguishable," "the clean subsample shows no signal") are equivalence claims wearing difference-test clothes. We therefore reassess every predictive-coefficient contrast as a two-one-sided-tests (TOST) equivalence assessment on the same day-cluster bootstrap machinery as Table 3, with $B = 9{,}999$ resamples of the trading-day clusters (fresh seed) and the bootstrap's 90% percentile interval as the test statistic: a contrast is equivalent at margin $\delta$ iff its entire 90% interval lies inside $(-\delta, +\delta)$. The primary margin follows replication practice —$\delta^* = 0.0074$, one half of the benchmark named coefficient, the "same order of magnitude" standard: a predictive difference below half the effect itself cannot reverse any qualitative statement about the instrument, while a difference above it would mean the two conditions measure materially different signals. In response units, $\delta^*$ is 27 bps of next-day return per one-standard-deviation firm-day sentiment shock (SD 0.361), against a benchmark response of 54 bps. Because a margin is a convention, each contrast's *equivalence margin threshold* (EMT) —the smallest margin at which it passes, i.e. $\max(|q_5|, |q_{95}|)$ —is reported alongside, so any reader-preferred margin can be tested by inspection. The margin rule was fixed before any verdict was computed; the analysis as a whole was designed in revision, after the difference tests were known, and is therefore labeled exploratory and reported outside the declared family (Section 3.4).

The table upgrades some conclusions and bounds others. *Upgraded*: all four named-versus-anonymized contrasts are formally equivalent at $\delta^*$ —EMTs between 0.0032 and 0.0062, i.e. 21–42% of the benchmark —with power 0.74–0.99 against a true difference of zero, so the de-identification invariance that Section 5.2 describes is an equivalence finding, not a failure to find a difference. *Bounded in both directions*: the swap contrast misses at $\delta^*$ (EMT 0.0083, 56% of the benchmark; power 0.75), so equivalence for the swapped instrument requires a wider margin than the convention provides; the dictionary comparison has an EMT of 4.4× the benchmark with zero power —an 80%-power equivalence test at $\delta^*$ would need roughly 165 times the present sample —so the LLM-versus-dictionary question is uninformative in both directions; and the priced-versus-clean contrast, whose difference-direction claim Section 5.4 reports as directional, fits no defensible equivalence margin either (EMT 6.6× the benchmark; its 90% interval sits entirely above zero, the same fact as the two-sided $p = 0.100$, i.e. one-sided

**Table 4:** Equivalence assessment (TOST) of predictive-coefficient contrasts

| Contrast | $\widehat{\Delta}$ | 90% CI | EMT | EMT/$\hat{\beta}$ | Power | Verdict at $\delta^*$ |
|---|---|---|---|---|---|---|
| Named − anonymized (cross-session) | +0.0004 | [−0.0035, +0.0040] | 0.0040 | 0.27 | 0.90 | Equivalent |
| Named − anonymized (same-session) | +0.0002 | [−0.0027, +0.0032] | 0.0032 | 0.21 | 0.99 | Equivalent |
| Named − anonymized (interleaved) | +0.0013 | [−0.0025, +0.0062] | 0.0062 | 0.42 | 0.74 | Equivalent |
| Named − anonymized (session 5) | −0.0018 | [−0.0052, +0.0021] | 0.0052 | 0.35 | 0.91 | Equivalent |
| Named − swapped (session 5) | −0.0039 | [−0.0083, +0.0004] | 0.0083 | 0.56 | 0.75 | Not established |
| Named − LM dictionary | −0.0119 | [−0.0658, +0.0415] | 0.0658 | 4.42 | 0.00 | Uninformative |
| Priced − clean (echo split) | +0.0436 | [+0.0005, +0.0976] | 0.0976 | 6.55 | 0.00 | Uninformative |
| Clean coefficient vs. zero | +0.0033 | [−0.0071, +0.0136] | 0.0136 | 0.91 | 0.00 | Not established |

$\widehat{\Delta}$: observed coefficient difference (next-day specification with controls; the echo rows use the same-day-control specification actually estimated). EMT: equivalence margin threshold, the smallest $\delta$ at which the percentile TOST passes, $\max(|q_5|, |q_{95}|)$ of the day-cluster bootstrap ($B$ = 9,999, seed 20261020). $\delta^*$ = 0.0074 (half the benchmark named coefficient; 27 bps per one-SD shock). "Power": analytic power at a true difference of zero (bootstrap SE, $z_{0.95}$ = 1.645). Sample multiples needed for 80% power at $\delta^*$: 0.5–1.1 for the four de-identification contrasts and the swap (the present sample meets or exceeds the requirement), 6.1 for clean-vs-zero, 143.7 for the echo split, 164.5 for the dictionary comparison. Exploratory, outside the declared family. Computed by `equivalence_testing.py`.

$p$ = 0.050). *Also bounded*: the clean subsample's coefficient is not established as equivalent to zero (EMT 0.0136, 91% of the benchmark; a sixfold sample would be needed), so "no clean signal" is as unsupported as "a clean signal half the baseline" —the honest statement is that this sample locates the loading in the echo subsample in point estimate only. Power is adequate exactly where the equivalence verdicts are positive: the current sample meets or exceeds what an 80%-power equivalence test at $\delta^*$ requires for the de-identification contrasts and the swap (0.5–1.1×) and falls short by two orders of magnitude for the dictionary and echo-split comparisons —the same asymmetry the power table documents for difference testing, now in the equivalence direction.

**Level-layer equivalence and an economics-anchored margin.** The battery above applies to prediction-layer coefficients; the level contrasts of Section 5.2 admit the same asymmetry between non-rejection and equivalence, so Table 5 re-runs the assessment on level differences in points at a pre-fixed margin of $\delta$ = 1.25 —half the two-and-a-half-point midpoint of the placeholder-rewrite family, fixed before inspection (`level_tost_and_margins.py`, seed 20261025, cluster bootstrap $B$ = 9,999).

**Table 5:** Level-layer equivalence (points; $n$ = 341 paired headlines, session 5)

| Level contrast | Mean diff. | 90% CI (headline) | EMT (headline) | Pass $\delta$=1.25 | EMT (14 clusters) |
|---|---|---|---|---|---|
| Named − swapped | +0.34 | [−1.06, +1.73] | 1.73 | No | 3.22 |
| Named − prefixed | −2.19 | [−3.50, −0.87] | 3.50 | No | 5.15 |
| Prefixed − anonymized | +0.13 | [−0.84, +1.10] | 1.10 | Yes | 2.39 |
| Swapped − anonymized | −2.39 | [−3.60, −1.18] | 3.60 | No | 4.53 |
| Named − anonymized | −2.05 | [−3.24, −0.86] | 3.24 | No | 4.35 |

EMT = the minimum margin at which the two-one-sided-tests verdict passes (the smaller the tighter the bound). The pre-fixed margin $\delta$ = 1.25 points is half the 2.5-point midpoint of the placeholder-rewrite effect family (2.05–3.13); margins are fixed before inspection and EMT is reported margin-free. Clustered EMT uses the 14-batch-call cluster bootstrap; under it no level contrast in this sample certifies equivalence below three points, and roughly 3.9× the headlines would be needed for 80% power at $\delta$ on the swap contrast headline-level. The rows with real displacement (swapped minus anonymized; named minus anonymized) fail as they should: they measure the de-identification effect itself. Computed by `level_tost_and_margins.py`.

Two readings. First, the swap-versus-named difference —the null that carries the no-material-prior conclusion —is small (13% of the placeholder midpoint) but not certifiable as equivalent at half the effect with 341 headlines; the honest statement in the main text is worded accordingly. Second, the economics-anchored margin: re-evaluating the prediction-layer battery at $\delta_{\text{econ}}$ = 10 bps of next-day response per one-standard-deviation sentiment shock —a round threshold independent of the benchmark coefficient,

whose $t$ ranges over $[1.47, 2.12]$ across draws —fails every contrast: minimum passing margins of 11.5, 14.5, 18.9, 22.3, and 29.8 bps/SD for the same-session, cross-session, session-5, interleaved, and swap contrasts, 48.9 for the clean-versus-zero bound, and 237 for the dictionary, with 3.6–7.8× the sample needed for 80% power on the identity contrasts. The equivalence verdicts of the main text are therefore margin-specific: this sample certifies instrument invariance at roughly a fifth to half of the benchmark effect, and no tighter.

# F Session Composition and Scoring-Session Drift

Table 6 documents the three original scoring sittings. The original named round scored all 542 headlines in two runs on 2026-10-15 (16:01 and 16:24 UTC); the audit round fielded the anonymized condition and three named repeats on the identical 341 in-window headlines in a single sitting on 2026-10-16 (04:24–04:33 UTC); the confirmatory session interleaved four named and four anonymized rounds plus the 542-composition control on 2026-10-18 (04:54–05:34 UTC, from the millisecond log). The named protocol is verbatim-identical across sittings (system prompt, user-message format, batch size 25; every request disables the model's extended-reasoning mode and otherwise uses the provider's default decoding parameters, as documented in Appendix A); the original batches additionally contained the 201 out-of-window headlines, so batch composition differs between sittings —and Appendix G measures exactly how much that difference is worth. A provenance distinction is recorded for the timestamps: the session-2 and session-3 times are read from millisecond-resolution entries in the batch logs (`llm_score_v2.log`, `llm_score_v3.log`), while the original scorer logged progress without timestamps, so the session-1 times are the modification timestamps of the first run's log file and of the archived output file (16:01:25 and 16:24:00 UTC respectively) —they mark the end of each run rather than its start, and are file-system evidence rather than in-log evidence; both record types are archived.

**Table 6:** Scoring sessions and cross-session stability of the named protocol

| Quantity | Value |
|---|---|
| Session 1 (named, 542 headlines) | 2026-10-15, 16:01–16:24 UTC |
| Session 2 (anonymized + 3 named repeats, 341 headlines) | 2026-10-16, 04:24–04:33 UTC |
| Session 3 (4 named + 4 anonymized interleaved, + 542-composition control) | 2026-10-18, 04:54–05:34 UTC |
| Gap, session 1 to session 2 | ≈ 12 hours |
| Named level drift, session 1 − session 2 (points) | +5.79 ($t$ = 7.03, $p < 10^{-11}$) |
| Same-session identity shift, named − anonymized (points) | −1.04 ($t$ = 1.60, $p$ = 0.111) |
| excluding the ten ticker-V headlines | −1.29 ($t$ = −2.05, $p$ = 0.041) |
| Identity shift, interleaved session 3 (points) | −2.82 (round-level $t$ = 6.8; clustered $t$ = 2.7) |
| excluding the ten ticker-V headlines | −3.13 (clustered $t$ = 4.8) |
| Batch-composition effect, within session 3 (points) | +3.61 (clustered $t$ = 2.7) |
| Session 4 replication, run order inverted (points) | +3.50 (clustered $t$ = 2.2) |
| Decomposition check | +5.79 − 1.04 = +4.75 (two-session shift) |
| Mean cross-session pairwise $r$ (original vs. repeats) | 0.905 |
| Mean within-session pairwise $r$ (repeats vs. repeats) | 0.950 |
| Mean \|diff\|: cross-session / within-session (points) | 11.3 / 6.5 |
| Anonymized-vs-repeat / repeat-vs-repeat \|diff\| (points) | 7.9 / 6.5 |
| ICC(2,3), named within session 2 | 0.949 |
| ICC(2,4), named within session 3 | 0.949 |
| ICC(2,4), pooled across sessions (published) | 0.919 |

Article level, $n$ = 341. Cross-session agreement is significantly worse than within-session agreement (Mann–Whitney on |diff| distributions, $p < 10^{-6}$): the scoring process varies between sittings by more than call-level noise alone. What looked like endpoint drift is now largely attributed: the batch-composition row shows that +3.61 of the +5.79-point cross-session gap is reproduced by composition alone inside a single sitting (Appendix G), with a non-monotonic endpoint residual of ±1 point per session pair. Rows computed by `revision_power.py` and `v_exclusion.py`; session-3 rows by `analysis_v3.py`; raw batch logs in `scripts/llm_score*.log`.

Three interpretive cautions. First, the drift is a level phenomenon: session 1's named scores are six points more bullish on average than session 2's named scores of the identical headlines, while the slope-relevant structure (which headlines are relatively positive) is largely preserved ($r = 0.905$ across sessions). Second, the pooled ICC(2,4)= 0.919 draws from both sittings and therefore already incorporates the cross-session component; within-session reliability is higher (0.949), and the interleaved session shows that within-session reliability is itself stable across sittings —0.949 in both. Third, the direction of endpoint drift is not a constant: the third sitting did drift back toward session-1 levels (+1.14 points at the 341 composition), exactly the caution the level anatomy implies. The honest constraint on pipelines remains a design rule rather than a correction: contrast conditions within a sitting, freeze the batch composition, and treat any cross-sitting comparison as confounded by construction.

*The naive two-session benchmark, archived in full.* The original contrast that Section 5.2 supersedes —named scores from 2026-10-15 (16:01–16:24 UTC) against anonymized scores fielded twelve hours later on 2026-10-16 (04:24–04:33 UTC), under a protocol verbatim-identical for the named conditions —read as a +4.75-point identity prior (paired $t = 5.14$, $p < 10^{-6}$): the out-of-sample analog of the "distraction effect" of Glasserman and Lin (2024). The surface pattern agreed with that reading —the two distributions correlate at 0.900 (Pearson; Spearman 0.888), 43.4% of headlines receive the identical score in both conditions, the mean absolute difference is 11.2 points, and within the October-16 sitting the anonymized-versus-named displacement (7.9 points of mean absolute difference) exceeds repeat-versus-repeat displacement (6.5 points) by roughly the one-point margin of the same-session level shift, while the cross-session displacement (11.3 points) towers over both. The session structure unravels it: the named protocol itself drifted +5.79 points between the two sittings (paired $t = 7.03$, $p < 10^{-11}$) —the same prompts, the same headlines, twelve hours apart, score six points more bullish in the first sitting —so the two-session shift decomposes exactly, +4.75 = +5.79 (session drift) −1.04 (same-session identity). Benchmarked within the October-16 sitting, the named-minus-anonymized level effect is −1.04 points —already negative in the named-minus-anonymized direction, with the mean test not reaching significance ($t = 1.60$, $p = 0.11$) while the rank test does (Wilcoxon $p = 0.04$; against the three repeats individually, +1.30, +1.67, and +0.15 in the anonymized direction, one of three below 0.05) —and the disclosure of a de-identification failure sharpens it: the executed alias table contained no "Visa" entry, so for the ten ticker-V headlines the anonymization was a no-op (Appendix B), and excluding them moves the same-session shift to −1.29 ($t = -2.05$, $p = 0.04$; Wilcoxon $p = 0.03$). Evidence of this shape supports a range, not a verdict: a same-session anonymization displacement of roughly one point, an order of magnitude below the session drift, whose significance depends on the test and the V exclusion. The interleaved estimate confirms the direction the same-session benchmark could only suggest, at roughly triple the magnitude (−2.82 versus −1.04), with the opposite sign from the naive +4.75 that session drift had masked entirely. The protocol lesson is general: condition contrasts fielded in separate sittings absorb session-level drift, and only same-session interleaving identifies the contrast the design intends.

Figure 2 shows why the naive reading was tempting —the two distributions overlap heavily and displace by an eye-catching five points —and Figure 3 displays the full level anatomy across the three sittings and two batch compositions, including the exact additive decomposition of the drift.

## G The Interleaved Replication Session

The confirmatory experiment was pre-specified in an archived design document before any session-3 data existed: re-field named and anonymized conditions on the identical 341 headlines *in one sitting*, with condition order alternated at the batch level, at least four rounds per condition, plus one pass of the identical headlines inside the original 542-article batch structure as a composition control. The executed design: four rounds; within each round, fourteen batches of 25; even-numbered batches call named-then-anonymized and odd-numbered call anonymized-then-named; eight score files plus the composition file; every call logged with a timestamp and model identifier (`glm-4-plus`); the analysis is `analysis_v3.py` and the raw log is `scripts/llm_score_v3.log`. Batch parity is a fixed property of a headline (its position in the stream), so the parity contrast identifies call-order contamination, not order effects: the

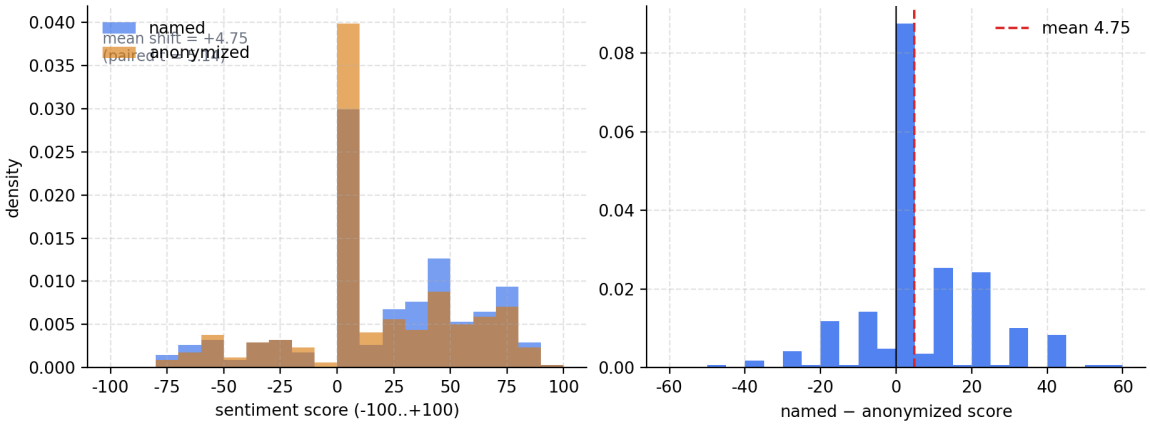



**Figure 2:** Named versus anonymized score distributions on the identical 341 headlines. Left: overlapping histograms (the named distribution sits to the right). Right: histogram of paired differences, with the mean shift of +4.75 marked. Both panels contrast the October-15 named round with the October-16 anonymized round and therefore mix identity with session drift; see Figure 3.

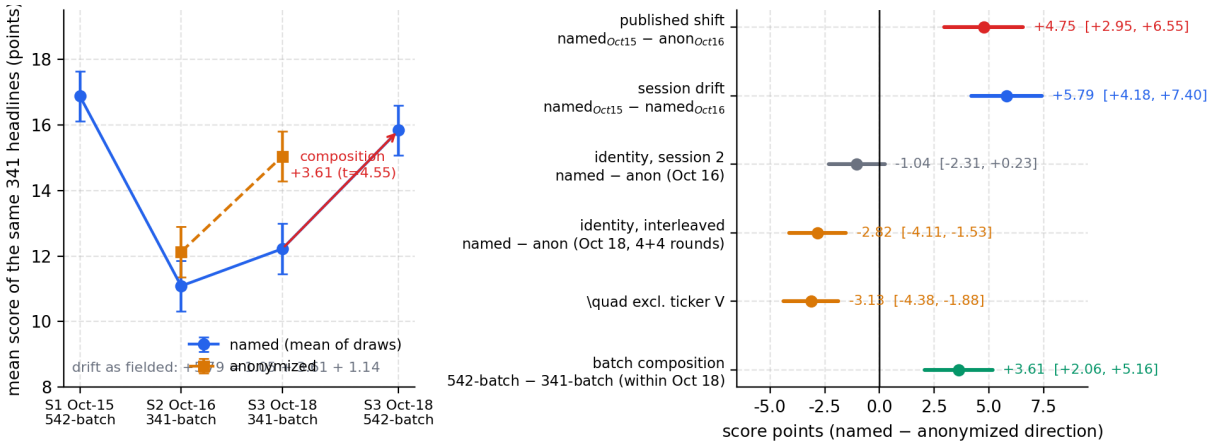



**Figure 3:** The level anatomy of the score across three sittings and two batch compositions. Left: mean score of the identical 341 headlines in each cell —the drift as originally fielded (+5.79, session 1 at the 542-batch composition to session 2 at the 341-batch composition) decomposes exactly into an endpoint leg (+1.05), the batch-composition leg (+3.61, measured within session 3), and the second endpoint leg (+1.14). Right: all named-versus-anonymized, drift, and composition contrasts with 95% CIs —the interleaved session establishes the de-identification effect at −2.82 where the non-interleaved session-2 contrast could only suggest it.

identity effect is 3.34 points (anonymized-minus-named) on named-first batches versus 2.27 on anon-first batches —no artifact.

The composition decomposition is exact by transitivity, and each leg is separately interpretable:

**Table 7:** Decomposing the cross-session named-level drift (points, article level, $n = 341$)

| Cell | Sitting | Mean score |
|---|---|---|
| Named, 542-article composition | 2026-10-15 (session 1) | +16.88 |
| Named, 341-only composition | 2026-10-16 (session 2) | +11.09 |
| Named, 341-only composition | 2026-10-18 (session 3) | +12.22 |
| Named, 542-article composition | 2026-10-18 (session 3) | +15.84 |
| *Legs of the decomposition* | | |
| Endpoint, session 1 − session 3 (both 542-composition) | | +1.05 |
| Composition, 542-composition − 341-composition (within session 3) | | +3.61 ($t = 4.55$) |
| Endpoint, session 3 − session 2 (both 341-composition) | | +1.14 |
| Total, session 1 (542) − session 2 (341) | | +5.79 |

The four cells identify the additive decomposition exactly: total drift = endpoint leg + composition leg + endpoint leg. The composition leg is measured inside a single sitting —same endpoint, same protocol, only the surrounding batch content differs —so re-batching alone accounts for 62% of what the two-session evidence had attributed to "the session." The endpoint legs are non-monotonic (session 3 sits between sessions 1 and 2), so the endpoint residual is a ±1-point-per-session-pair wander, not a trend. Computed by `analysis_v3.py` from `llm_scores_s3_*.json` and the archived session-1/2 scores.

Anonymized levels track the same structure: +12.13 (session 2) and +15.04 (session 3, four-round average) —the anonymized protocol moves between sittings too, which is why cross-session named-versus-anonymized contrasts absorb drift on *both* sides.

**The mechanism sitting (2026-10-18, fourth session).** A fourth sitting ran the pre-specified discriminating design, with the run order *inverted* so that time-of-session works against the composition effect: (i) the 542-article composition pass ran first, (ii) the standard 341 pass second, and (iii) a seeded random shuffle of the 341 third (seed 20261018, order archived as `s4_shuffle_order.json`; the original order is company-blocked by collection, the shuffle mixes companies within every batch). Three results. First, the composition effect replicates under the inverted order at +3.50 (batch-call-clustered $t = 2.17$, $p = 0.042$; pooled with session 3: +3.56, $z = 3.4$): since the 542 pass ran *early* this time, a time-of-session explanation would have flipped the sign —the effect is batch composition, not sitting position, and the drift between sessions 3 and 4 at fixed composition is +0.56 ($t = 0.3$). Second, re-batching the *same* 341 headlines at random moves levels by +4.72 (clustered $t = 2.02$, $p = 0.065$ —suggestive, one sitting): batch assignment matters even within a fixed corpus, so the composition channel is not specifically about out-of-window headlines. Third, the mechanism:

The mechanism table carries the paper's most cautionary arithmetic, and the accounting fails in both directions at once. A naive reading —"anchoring $\theta = 0.36$, batch means differ, therefore levels shift" —is barred for re-batching a fixed corpus by the telescoping identity (linear anchoring redistributes scores *within* a corpus but provably cannot move the average), and for the composition contrast, where the mate pool genuinely changes, the linear part is computable and negligible: the out-of-window titles that join the 542-article batches are, if anything, slightly more negative than the headlines they sit beside, so $\theta$ predicts +0.04 of the observed +3.56, and closing the gap linearly would require a +30-point content gap against an actual −1.3. The estimated asymmetric assimilation does not close it either, because the added mates are less positive than those they replace. The honest statement is layered: the *existence* of the batch channel is established at $z \approx 3$–5 across two sittings with inverted run order; its *linear* component is bounded below 1% of the estimate; its precise non-linear mechanism —among contrast-type responses to batch context, within-response position, and the response-level mood —remains open. The practical rule is unaffected by the open derivation: hold the batch composition fixed across anything being compared, because re-batching moves levels by more than most effects of interest, and the direction of the move is not predictable from the batch means alone.

**Table 8:** Mechanism statistics (eight named-condition passes, sessions 3–4)

| Quantity | Estimate |
|---|---|
| Clean anchoring on batch-mates' fixed content, $\theta$ | +0.36 ($t$ = 4.9); full mate pool: +0.38 ($t$ = 5.0) |
| asymmetry: extra slope when batch-mates' mean is positive | +0.69 ($t$ = 2.3; negative-mate slope ≈ 0) |
| Response-level mood (shared by the 25 scores of one API response) | SD 2.6; 13% of residual variance |
| Idiosyncratic per-headline noise | SD 6.7; 87% |
| Level shift predicted by *linear* anchoring, re-batching a fixed corpus | exactly 0 (telescoping identity) |
| Level shift predicted by *linear* anchoring, composition contrast | +0.04 of the observed +3.56 (≈1%) |
| out-of-window mates ($n$ = 201) vs. in-window ($n$ = 341): grand mean | +12.60 vs. +13.91 (gap −1.3; 35% vs. 62% positive) |
| content gap needed for a full linear explanation / actual gap | +30.1 / −1.3 (wrong sign) |
| Observed level shifts: 542-composition / random re-batch | +3.56 pooled ($z$ = 3.4) / +4.72 ($p$ = 0.065) |

The anchoring coefficient regresses each headline's score on the mean of its batch-mates' grand means (estimated across all passes, noise-averaged), with headline and pass fixed effects —the regressor is a deterministic function of the batch assignment, free of same-response noise endogeneity. The telescoping identity: for any partition into balanced batches, the average of *leave-one-out batch-mate means* over all headlines equals the global mean, so a linear anchoring term cannot move the corpus mean of a re-batched *fixed* corpus. The composition contrast is the one case where the mate pool itself changes, and there the linear prediction is computable rather than zero: with the full mate pool (in-window and out-of-window batch-mates), the mean leave-one-out mate content moves by +0.10 points between the two compositions, so $\theta \approx 0.36$–$0.38$ predicts +0.04 —about 1% of the observed +3.56 —and a full linear explanation would need the 201 out-of-window titles to average +30 points *above* the in-window ones when they in fact average 1.3 points below. The estimated asymmetric assimilation does not close the residual either (the added mates are *less* positive, so asymmetric assimilation alone predicts a small decrease). What remains is the class of response-level, non-linear components the design does not separately identify —contrast-type responses to batch context, within-response position, and the response-level mood —so the *existence* of the batch channel is established at $z \approx 3$–$5$ across two sessions with inverted run order, its linear component is bounded below 1% of the estimate, and its precise non-linear mechanism is open. Computed by `anchoring_regression3.py`, `reinference.py`, and `batch_mechanism_quant.py`.

**Downstream consequences of the composition effect.** If levels move but slopes do not, what is the harm? Table 9 and Figure 4 quantify which downstream uses of the score survive re-batching and which do not, using the archived sources of sessions 1–4 (all named protocol, identical 341 headlines; staged addition computed from archived draws, outside the declared family; `batch_downstream.py`, seed 20261019). Slopes and ranks are robust: the predictive coefficient stays in $[0.010, 0.016]$ across all seven sources, every composition and re-batch contrast of slopes is insignificant ($p > 0.9$), and the cross-sectional ranking of firm-days correlates at $\rho \geq 0.81$ across compositions (0.97 at fixed composition across sessions). Absolute-scale uses are not robust: between the two compositions of the same session, a fixed "score ≥ +50" filter more than doubles its trade count (22 to 50 firm-days in session 3; 27 to 43 in session 4), the daily-mean series shifts by 4.0–4.6 points, and the sign of the day-over-day change in daily sentiment flips on 16–24% of trading days. The composition effect is a *level* pathology with boundary costs: harmless to conditional, cross-sectional, and regression uses in this sample, and disqualifying for threshold rules, absolute-score comparisons across pipelines, and time-series signals built on the score's level.

# H Session-5 Controls: Archived Design Document and Results

A fifth sitting —identity-swap (each headline's focal identifiers replaced by a seeded random other firm's identifiers, so the text remains a natural company headline but names the wrong company), neutral prefix ("[Company X]" over the corrected anonymized text), corrected anonymization (the alias table now covers the ticker-V identifiers of Appendix B; a forced verification pass reported zero residual focal identifiers across all transformed texts), and two named passes at `temperature = 0` —was pre-specified, with falsifiable predictions, in an archived design document deposited before any of its data was collected (`audits/P2_preregistration_session5.md`); per Section 3.4 this is within-study staged pre-specification, not independent pre-registration, and we do not represent it as the latter. The design: six conditions over the identical 341 headlines in a single sitting (2026-10-19), batches of 25, six-way ro-

**Table 9:** What re-batching does to downstream uses of the score (archived sources, sessions 1–4)

| Downstream use | 341-composition | 542-composition / re-batch | Verdict |
|---|---|---|---|
| Predictive slope $\beta$ (session 3) | 0.0146 ($t$ = 1.67) | 0.0128 ($t$ = 1.96); diff $p$ = 0.97 | robust |
| Predictive slope $\beta$ (session 4) | 0.0157 ($t$ = 1.49) | 0.0110 / 0.0104; diffs $p > 0.9$ | robust |
| Firm-day rank stability (Spearman $\rho$) | — | 0.91 / 0.86 / 0.81 (re-batch) | robust |
| Q5−Q1 spread (bps, Welch $t$) | +156 (1.5) | +95 (1.2) / −2 (−0.1) | unstable, always insignificant |
| Firm-days ≥ +50 (session 3) | 22 (10.7%) | 50 (24.4%) | *doubled* |
| Firm-days ≥ +50 (session 4) | 27 (13.2%) | 43 (21.0%) / 49 (23.9%, re-batch) | *doubled* |
| Daily-mean level (points) | reference | +4.5 / +4.0 / +4.6 | shifted |
| Day-over-day sign flips of daily mean | reference | 16% / 17% / 24% | contaminated |

All rows from `batch_downstream.py` (seed 20261019) over the identical 341 in-window headlines scored under the named protocol: session-3 pair (341-only four-round average vs. the same session's 542-composition pass) and session-4 triple (341-only pass vs. the 542-composition pass vs. the seeded random re-batch). Slope differences use day-cluster bootstrap ($B$ = 999). Spread columns report the Q5−Q1 mean next-day return with Welch $t$ in parentheses. The threshold rows count firm-days whose mean score is at least +50 points; the daily-mean rows describe the daily cross-sectional mean of firm-day sentiment. "Robust" means the contrast across compositions is statistically zero; threshold and time-series uses move by construction because the level moves.

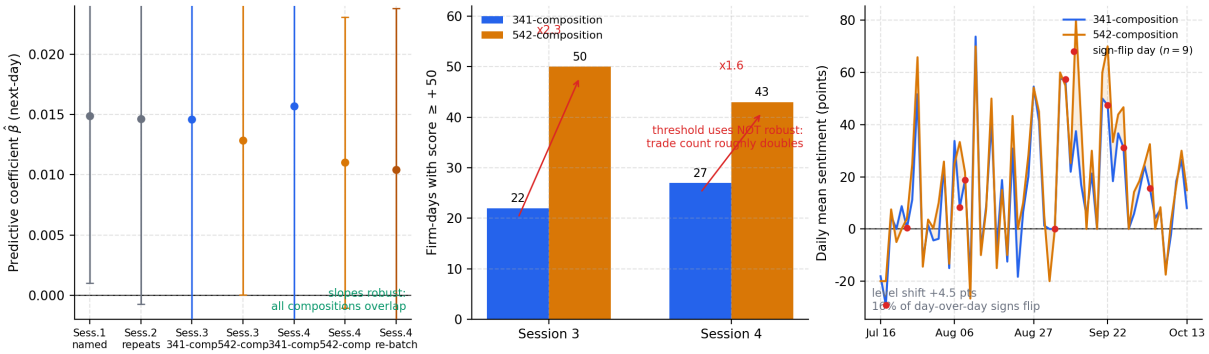



**Figure 4:** The downstream consequences of batch composition at a glance (companion to Table 9). Left: the next-day predictive coefficient across all seven archived named-protocol sources with 95% CIs —341-composition, 542-composition, and re-batched sources (colors) —overlap everywhere: conditional and cross-sectional uses are robust. Middle: firm-days crossing the fixed "score ≥ +50" threshold under the two compositions of the same session —the trade count roughly doubles (22 to 50; 27 to 43): absolute-threshold uses are not robust. Right: the daily cross-sectional mean sentiment of the identical firm-days under the two compositions of session 3 —a +4.5-point level shift, with days whose day-over-day change flips sign between the two compositions marked (9 of 58 changes, 16%): time-series uses are contaminated. Computed by `figures_v7.py` from `batch_downstream.py` outputs and the archived session-3 scores.

tated call order per batch so every condition occupies every time-of-session position uniformly, protocol verbatim-identical to sessions 1–4, every call logged with timestamps and served-model identification. Collection was interrupted by sustained provider rate-limiting after 63 of the 84 planned calls and completed later the same UTC day under the identical archived protocol (84 calls total; the interruption, the resume-safe checkpoints, and the per-call log are all in the replication package, and the sealed partial score files are hash-bound in `aspredicted_session5_form.md`); the analysis below ran only after completion, under the locked plan (`analysis_v5.py`, seed 20261019).

The pre-specified predictions and their outcomes:

**Table 10:** Session-5 controls: pre-specified predictions and outcomes (points on the $[-100, 100]$ scale; $n = 341$ paired headlines)

| Contrast | Prediction | Outcome |
|---|---|---|
| P1: named − anonymized-v2 | $\approx -2.8$ | $-2.05$ ($t = -2.84$; Wilcoxon $p < 0.001$; clustered $p = 0.18$) |
| excl. ticker V | | $-2.15$ ($t = -2.89$) |
| P2: named − swapped | firm prior: large; perturbation: $\approx 0$ | +0.34 ($t = 0.40$; clustered $p = 0.86$) |
| swapped − anonymized-v2 | | $-2.39$ ($t = -3.26$; clustered $p = 0.08$) |
| P3: prefixed − anonymized-v2 | $\approx 0$ | +0.13 (clustered $p = 0.92$) |
| P4: temp-0 round *a* vs. round *b* | sampling noise: bitwise identical | 83.3% identical; within-SD 4.8 (vs. 7.2 default) |
| P5: temp-0 level vs. named default | no prediction (two-sided) | +1.06 ($p = 0.052$) |
| *Prediction layer (H1 coefficient per condition, day-cluster differences)* | | |
| $\beta$, named / anonymized-v2 / swapped / prefixed | no difference | 0.0140 / 0.0158 / 0.0179 / 0.0134; |
| named − anonymized-v2 / named − swapped | | $p = 0.93$ / $p = 0.92$ |

Headline-level paired $t$ with batch-call-clustered inference (14 clusters) as the conservative default, exactly as pre-specified and as in session 3. P1 is direction- and magnitude-consistent with the interleaved de-identification effect at $-2.05$ (against session 3's $-2.82$) with the ten ticker-V headlines now genuinely anonymized. P2 is the disambiguating contrast: a firm-specific prior requires the swapped condition (natural text, wrong identity) to load onto the wrong firm's prior and diverge from named; the observed +0.34 —far inside the repeat-noise floor of 7.2 —provides no support for that reading, and the swapped condition sits with the named family ($-2.39$ below anonymized). P3 clears the prefix format. P4: temperature zero tightens the instrument (within-SD 4.8) but does not make it deterministic —57 of 341 headlines still move between two identical temperature-zero calls, a pattern consistent with server-side nondeterminism. Computed by `analysis_v5.py` from the six archived session-5 score files.

The outcome pattern re-attributes the published level channel: the named-versus-anonymized displacement is best read as a *de-identification* effect (the removal of a concrete identifier), with no support for a firm-specific prior —the wrong firm's name sits with the right firm's name and two-plus points from the placeholder —and the paper adopts the accurate name throughout. The temperature-zero result strengthens the stochasticity claim beyond decoding settings, and the prediction layer confirms that no identity variant moves predictive content.

# I Session-6 Controls: Fictional Names and the Batch-Size Dose-Response

Two further staged sittings (2026-10-25), pre-specified in an archived design document before any of their data was collected (`audits/P3_preregistration_session6.md`); per Section 3.4 these are within-study staged pre-specifications, not independent registrations, and both are reported outside the declared family.

**The fictional-name control.** The session-5 swap control pairs natural text with a *different real* firm's identity; the missing cell is a name that is concrete but carries no real-world prior. The sixth sitting therefore fields three conditions over the identical 341 headlines in a single sitting, three-way rotated call order, protocol verbatim-identical: a fresh named anchor (N6), a fresh anonymized-v2 round (A6), and a fictional-name round (F6) whose focal identifiers are replaced by invented firm names (`s6_fictional_map.json`: thirty-six two-part invented names with synthetic tickers, constructed to read naturally in headlines and to carry no association with any prominent public company; any incidental collision with an obscure real name is disclosed as a construction limitation). The pre-collection verification gate blocked the first attempt: six F6 texts retained their original identity through concatenated name

variants ("ExxonMobil," "JPMorganChase") that the session-5 word-boundary check cannot see —the same failure class as the ticker-V anonymization gap of Appendix B, caught this time before any call was made. The session-6 transform adds concatenated-variant replacement; the re-run gate reports zero residuals on all 341 texts across both tiers (word-boundary and substring), zero missing fictional names, and zero anonymization residuals (`s6_verify_report.json`); the first-attempt block is archived in `llm_score_v6.log` as a design deviation.

**Table 11:** Session-6 controls: pre-specified predictions and outcomes (points; $n = 341$ paired headlines)

| Contrast | Prediction | Outcome |
|---|---|---|
| P6a: fictional − named | rewrite hyp.: ≈ 0; identity hyp.: +2 to +3 | +1.20 ($t = 1.64$; clustered $p = 0.45$) |
| P6a: fictional − anonymized | rewrite hyp.: +2 to +3; identity hyp.: ≈ 0 | −1.33 ($t = -1.72$; clustered $p = 0.45$) |
| P6b: named − anonymized | ≈ −2 (replication cell) | −2.54 ($t = -2.92$; Wilcoxon $p = 0.002$; clustered $p = 0.15$) |
| P6c: $\beta$ named / anonymized / fictional | no difference | 0.0126 / 0.0112 / 0.0113; all pairwise $p \geq 0.62$ |

Headline-level paired $t$ with 14-cluster clustered inference, as in sessions 3–5. Neither P6a sub-contrast is individually established under clustering; the joint pattern —the fictional level at 13.90, almost exactly midway between named 12.70 and anonymized 15.23 —is what separates the pure hypotheses: the placeholder-rewrite reading predicted F6 with N6, the identity-removal reading predicted F6 with A6, and the data place it between, indicating a roughly even split between a real-name component and a rewrite component. Level-layer equivalence margins for these contrasts: EMT 2.41 points headline-level (F6−N6) and 2.62 (F6−A6), against the 1.25-point margin of Table 5. The two clustered $p$-values are near-identical (0.4505 and 0.4502) not by transcription but by construction: the fictional level sits almost exactly midway, so both sub-contrasts share clustered $|t| \approx 0.755$ (`results_s6.json`). Computed by `analysis_v6.py` (seed 20261025).

**The swapped-text leakage audit.** The swap transform replaces focal identifiers only; product, platform, and executive names survive it. `swap_leakage_audit.py` rebuilds all 341 swapped texts and flags them with a transparent, data-driven cue lexicon (a token is a firm cue if it appears in at least three of that firm's in-window titles and in no more than five other firms' titles, excluding the firm's own names and a finance-generic stoplist; a broader rule drops both thresholds). Results: 51 of 341 swapped headlines (15.0%) still contain at least one original-firm lexical cue under the primary rule (91, 26.7%, under the broad rule), and six (1.8%) retain the original identity outright through the concatenated name variants above —the audit that caught what the verification gate had missed. Excluding every flagged headline leaves the swap contrast at +0.36 (primary rule) to +0.42 (broad rule) against the full-sample +0.34, moves the swapped-minus-placeholder gap to −2.64/−2.52 (larger in magnitude), and leaves the de-identification contrast at −2.28/−2.10 —the swap null is not an artifact of identity leakage, and if anything the leakage was diluting the placeholder-versus-swap separation. The lexicon is a lexical heuristic, reproducible from the archived corpus, and is not human annotation (`s6_cue_lexicon.json`, `results_swap_leakage.json`).

**The batch-size dose-response.** A seventh pass-set (same sitting day, pre-registered as session 6b) scores a seeded 50-headline subsample (seed 20261025, order archived) alone, in fives, and in twenty-fives —one pass per size, singleton calls split around the other passes so the size-1 level is centered in session time. One wide null, one anomaly, one underpowered decomposition. First, the singleton contrast is a wide null rather than a zero point: the singleton-minus-25 difference is −0.20 points (paired $t = -0.09$, $n = 50$; SE 2.35), so its 95% CI of $[-4.92, +4.52]$ points cannot exclude a shift of the +3.6-point composition effect's own size, and an equivalence test would certify the singleton contrast only at a margin of 4.1 points —wider than the effect itself. The dose-response therefore cannot rule batch-mate presence out as a level channel: what the composition sessions establish is that *changed* mate composition moves levels, and whether presence alone also does is left open. Second, the suggestive anomaly: the size-5 pass sits 5.5–5.7 points below both neighbors ($t$ between 2.3 and 2.5); with one pass per size this is a single-session reading carried by ten calls, it exceeds the documented within-sitting drift scale, and we report it as unexplained rather than interpret it. Third, the pre-registered noise decomposition is underpowered at this design: the mood-variance estimator has a bootstrap interval that spans the session-3 benchmark (mood SD ≈ 2.6 points), and the triangle solution of the three pairwise per-title variances gives per-call

noise SDs of 12.3, 11.6, and 11.2 points at sizes 1, 5, and 25 —all above the session-3 within-round benchmark of 7.2, consistent with the serving-stack noise varying across days (Section 5.3), but with standard errors that cannot separate the size-25 minus size-1 difference (a $\approx$ 6.8-point$^2$ benchmark against $\approx$ 56-point$^2$ sampling noise). Resolving the mood-versus-position decomposition needs multiple passes per size, which we leave to future work with the design archived. Computed by `analysis_v6b.py` from `llm_scores_s6b_{b1a,b25,b5,b1b}.json` (`results_s6b.json`).

**Cross-session pooled estimate of the de-identification effect.** Each session's estimate is individually fragile under call-clustered inference —session 3: $-2.82$ (clustered $t = 2.7$, $p = 0.017$); session 5: $-2.05$ ($t = -1.36$, $p = 0.18$); session 6: $-2.54$ ($t = -1.43$, $p = 0.15$) —so the main text reports the pooled estimate as its summary number. A DerSimonian–Laird random-effects meta-analysis on the three clustered estimates (SEs 1.04, 1.51, and 1.78) gives $-2.57$ points with 95% CI $[-4.08, -1.06]$ ($z = 3.34$; $Q = 0.18$ on 2 df, $\tau^2 = 0$, $I^2 = 0\%$ —no detectable between-session heterogeneity). Because the three sessions re-score the same 341 headlines, the estimates are not independent draws: the per-headline anonymized-minus-named responses correlate across sessions (Pearson $\rho = 0.45$ for sessions 3–5, 0.40 for 3–6, 0.32 for 5–6), so the de-identification response is headline-specific and persistent rather than per-call noise —and, less conveniently, the pooled interval that treats sessions as independent is too narrow. A conservative variant that adds the corresponding covariance terms to the full clustered standard errors gives 95% CI $[-4.56, -0.58]$, still excluding zero. Session 3's estimate retains the ten ticker-V headlines whose anonymization was a no-op; its V-excluded variant is larger in magnitude ($-3.13$, headline-level $t = 4.8$), so the pooled figure is, if anything, conservative. Computed by `meta_deident_pooling.py` from the archived session score files (`results_meta_deident.json`).

# J Classification Audit of the Price-Language Flag

All 52 headlines flagged by the realized-price regular expression of Section 5.4 were re-classified one by one against a written rulebook with four categories: **P** —the headline states or clearly implies the company's stock price has already moved (including moves stated with a fundamental cause, e.g. "shares jump on blowout results"); **E** —the matched keyword is a fundamental magnitude ("revenue rises 12%"), with no realized stock move; **D** —a directional forecast only ("could sink," "may drop to $300"); **N** —a regex false positive ("execution short fall"; "Walmart just dropped October deals" —retail promotions). The tie rule: dual-content headlines with both a realized move and a fundamental result classify as P, since the defining feature of an echo headline is that the move is reported. The classifier is disclosed for what it is: LLM-assisted, structured, item-by-item, run by the same class of model that produced the scores —an internal consistency audit of the flag, not independent human annotation, and not a gold standard (the rulebook and all 52 labels are archived as `priced_classification.json`). Human two-rater labels against the rulebook —and a sampled check of non-flagged headlines for the classifier's false-negative rate —are the natural validation and remain outstanding; the machine labels above are the only audit of the flag this draft can report, and the split's directional status ($p = 0.10$) prices that caveat.

# K A Minimal Viable Audit Protocol

The protocol below is the paper's working answer to "what should a small-sample LLM-sentiment study run before reporting a predictability statistic?" It is deliberately minimal: every component costs at most one additional scoring pass over the study's own corpus, and the whole protocol is executable in an afternoon at headline-scale prices. Each item names the failure mode it guards against, as documented somewhere in this paper.

1. **Anonymize once, and verify the anonymization.** Replace firm names and tickers with a neutral placeholder, then run a forced residual-identifier check over every transformed text. Guards against

**Table 12:** Auditing the echo classifier and the echo split

| Quantity | Value |
|---|---|
| Flagged headlines / re-classified as realized-price (P) | 52 / 43 (82.7%) |
| Fundamental magnitude (E) / directional forecast (D) / false positive (N) | 3 / 3 / 3 |
| Firm-days, regex flag vs. strict-P | 38 vs. 33 (20 companies) |
| $\beta$, regex-flagged vs. strict-P (same-day control spec) | 0.0470 ($t$ = 1.80) vs. 0.0475 ($t$ = 1.61) |
| Mean \|same-day return\|, strict-P vs. clean vs. flagged-non-P | 361 / 178 / 120 bps |
| Largest single-firm share of strict-P firm-days | 9% |
| Leave-one-firm-day-out: $\beta$ range / sign flips of diff | [0.018, 0.059] / 0 |
| Leave-one-company-out: $\beta$ range / sign flips of diff | [0.019, 0.056] / 0 |

The refined split uses only firm-days with at least one strict-P headline. The misclassified firm-days' 120 bps of same-day absolute return —below even the clean group's 178 —is the strongest internal validation of the classification: headlines that are not echoes do not behave like echoes. Jackknife rows drop each strict-P firm-day (or each company holding one) and re-estimate both split coefficients; the difference never changes sign in any replicate. Computed by `p11_classification_analysis.py` from `priced_headlines.json`, `priced_classification.json`, and the archived panel.

identity priors, and against silent no-op anonymizations (the ticker-V failure of Appendix B: ten headlines whose "anonymized" text still named the firm).

2. **Field the named-versus-anonymized contrast interleaved in a single sitting**, alternating condition order at the batch level. Guards against cross-session level drift masquerading as a treatment effect (+5.79 points over twelve hours here; Section 5.2).
3. **Freeze the batch composition** across every comparison —conditions, sessions, or model versions —and re-score once under an alternative composition to measure the channel on your own corpus. Guards against batch-composition level shifts (+3.6 points here; harmless to slopes and ranks, disqualifying for thresholds and time series; Appendix G).
4. **Score twice and report the noise floor:** an ICC or within-headline repeat SD alongside any single-call statistic, with inference clustered at the call level. Guards against response-level mood and overstated precision (ICC 0.92 here; 13% of residual variance is shared within a response; Section 5.3).
5. **Flag realized-price language and audit the flag** against a written rulebook, on every flagged headline if the count is small, on a sample otherwise. Guards against retrospective-coverage contamination (the echo channel of Section 5.4: the predictive loading directionally sits in the 15% of headlines that quote realized moves).
6. **Test the sample for selection:** a two-line firm-and-day fixed-effects regression of realized activity on the coverage indicator. Guards against event-sampling bounds on external validity (coverage selected on volatility at $t$ = 2.76 here; Section 5.5).
7. **Declare the test family before the augmented inference, and compare instruments by difference tests, never by separate significances.** Guards against multiplicity and the genre's most common inference error (Sections 3.4 and 7).

Items 1–4 protect the measurement; items 5–6 protect the interpretation of the return regression; item 7 protects the inference. A study that cannot run the full protocol should at minimum report which items it skipped, so that readers can price the corresponding channels themselves.

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